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Measuring subjective meaning structures by the laddering method: Theoretical considerations and methodological problems

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Abstract

Starting from a general model of measuring cognitive structures for predicting consumer behaviour, we discuss laddering as a possible method to obtain estimates of consumption-relevant cognitive structures which will have predictive validity. Four criteria for valid measurement are derived and applied, which refer to data collection, coding, and analysis. These criteria are evaluated using examples from a laddering study where additional data was collected to shed light on respondents' subjective interpretation of the laddering task as well as interviewers' experiences with it. Several possible validity threats are identified, and ways to improve the method are suggested.

Keywords: Cognitive structure; Means-end chains; Laddering; Interview techniques

1. Introduction

Means-end chains (MEC) and the laddering technique as a way to measure them have become popular in consumer research. But what is actually being measured, what is the purpose of this type of research, and what do we know about the validity of the results? Answers to these questions could help improve the measurement methodologies, and could stimulate more theory development in the area. The purpose of this paper is to make a first step in this direction. We start by using a cognitive framework to come up with four criteria for the validity of results obtained by the laddering method. For each of the four criteria, we then identify a number of problems in laddering research practice, and make suggestions for how they can be solved.

2. Criteria for the validity of laddering results

2.1. What is laddering supposed to do?

Any judgement of the validity of the results of a data collection device depends, of course, on what the data collection device is supposed to achieve. The literature in MEC and laddering is not very explicit about this. We believe that two basic views can be distinguished. For the sake of simplicity, we will call them the *motivational* and the *cognitive structure* view.

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The *motivational view* is that laddering and MEC are concerned with obtaining insight into consumers' buying motives –i.e., in the way basic motives are linked to shopping behaviour. Laddering and MEC would then be a modern variant of motivation research in the Dichter (1960) tradition. Laddering can give valuable insights by prompting consumers to reflect on their buying motives in a way not typical for daily shopping behaviour. Such insight is necessarily qualitative in its character, and the kind of structures derived will be situationally constructed meanings. The main criteria for evaluating the usefulness of the approach would be to which extent users of the results feel that they have achieved a better understanding of consumers that gives inspiration to managers and helps them make better business decisions.

The *cognitive structure view* is that MEC are a model of consumers' consumption-relevant cognitive structure, i.e., of the way consumption-relevant knowledge is stored and organized in human memory. A basic hierarchical model is assumed, in which cognitive categories of different levels of abstraction are interlinked in chains and networks, and where it is assumed that behavioural motivation is derived by linking cognitive categories corresponding to concrete objects –usually products –with cognitive categories at a high degree of abstraction, like values. It should then be possible to explain and/or predict actual behaviour with regard to these concrete objects by specifying how, in a given situation, parts of the cognitive structure are retrieved and used to guide behaviour. The main criterion for evaluating the usefulness of the approach would be the predictive ability of the estimates of the cognitive structure obtained, when used as input in a theory which explains behaviour by how, in a given situation, information is retrieved from the cognitive structure and used in decision-making.

The literature on MEC and laddering does not take a clear stand on which of these two views are being endorsed. The Grey Benefit Chain (Young and Feigin, 1975), one of the early approaches, seems to be most in the motivation research tradition. Asselbergs (1989), while adapting schema theory and therefore a cognitive perspective, regards MEC generated by laddering not as measures of cognitive structure, but as a 'reconstruction' of relevant information, which also tends towards the first view. Gutman, Olson, and Reynolds (Gutman, 1982; Jolly et al., 1988; Olson, 1989; Olson and Reynolds, 1983; Peter and Olson, 1990; Reynolds and Gutman, 1988) adopt a cognitive structure perspective, saying that the hierarchical value map derived from laddering data is "an aggregate map of cognitive structure" (Olson and Reynolds, 1983).

Of course, the choice depends on which general view of scientific analysis in consumer research one adopts. If one adopts the phenomenological view that all measurements are context-dependent, the notion of measuring a situation-invariant cognitive structure does not make much sense. If one, on the other hand, adopts a critical rationalism perspective as would be typical for most of cognitive psychology, measuring only situationally-dependent meaning would appear unsatisfactory.

In the following, we adopt the second view –the view that MEC and laddering aim at measuring consumers' consumption-relevant cognitive structures. This implies that the question of the validity of the results obtained can be treated in analogy to classical test theory –i.e., as the correspondence between measured value and 'true' value. Specifically, we would examine the correspondence between a respondent's (or a group of respondents') 'true' cognitive structure and the estimate of cognitive structure obtained by laddering. The problem is, of course, that we have no external criterion for a respondent's 'true' cognitive structure, since cognitive structure is always mediated by cognitive processes, before it determines behaviour. This problem is addressed in the next section.

2.2. *The relationship between cognitive structures, processes, and behaviour*

The cognitive view of human beings regards behaviour as being directed by *cognitive structures* and *cognitive processes*. A cognitive structure is the organization of experience and other types of information in human memory. Cognitive struc-

tures are usually modelled as a network of cognitive categories and the associations between them or as schemas related by inference processes. A system of MEC can be regarded as an *excerpt* from the cognitive structure that is considered relevant for explaining buying behaviour. MEC map only those cognitive categories which are directly or indirectly linked by associations of a special type, namely subjective causality, to cognitive categories corresponding to concrete product attributes (Gutman, 1991). In the typical laddering task, the researcher attempts to restrict the generation of concrete product attributes to salient choice criteria, which increases the likelihood of their relevance for buying behaviour.

Cognitive processes are the processes by which the cognitive structures are changed due to new information from the environment, and by which information is retrieved from the cognitive structures and used to direct behaviour. Two types are usually distinguished. *Automatic* cognitive processes are unconscious, not subject to capacity

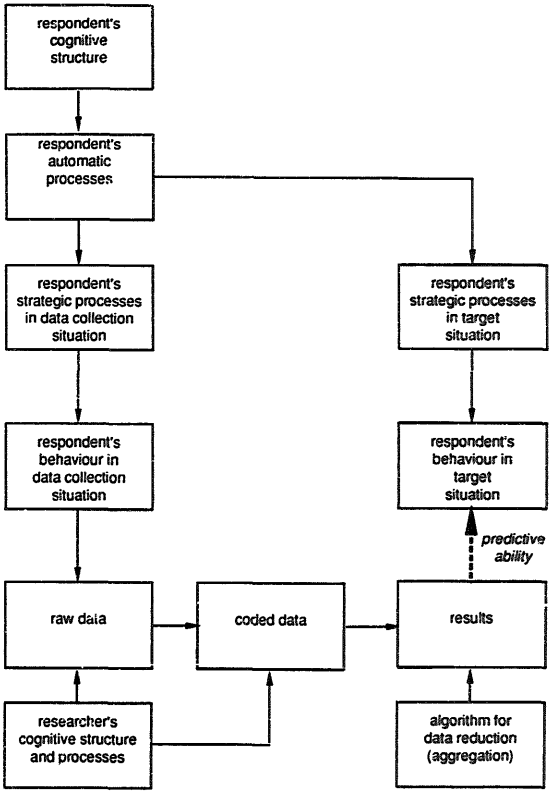


Fig. 1. A general cognitive model for data collection, analysis, and prediction.

limitations, and quite invariant with regard to different task requirements. They require a large amount of learning, but are very efficient once learned. Perceiving familiar stimuli in the environment and linking them to cognitive categories stored in memory, and retrieving information from memory based on retrieval cues are examples for automatic cognitive processes. *Strategic* cognitive processes are conscious, subject to capacity limitations, and easily adapted to task requirements. They refer to intentional, problem-solving-like mental activities, like trying to find meaning in unfamiliar stimuli, or combining retrieved information in new ways to arrive at evaluations or decisions (K.G. Grunert, 1991; Schneider and Shiffrin, 1977; Strube, 1984).

Human behaviour is the result of an interaction of cognitive structures and cognitive processes. This is true for buying behaviour, and also for behaviour in a data collection situation, such as a laddering task. This has implications for the evaluation of the validity of the results from laddering. It means that it is impossible to obtain a 'pure' measure of cognitive structure, because the behaviour of a respondent in a data collection situation is a result of, and therefore an estimate of, both cognitive structure *and* cognitive processes. The best we can do, then, is to aim for an estimate of cognitive structure *and* cognitive processes which will be the most useful in explaining/predicting the kind of behaviour we are interested in. In the context of laddering, this may be broadly defined as all behaviour directed towards or triggered by the concrete objects constituting the starting point for the means-end chains – namely, products and their attributes. This encompasses product class choice, brand choice, information acquisition behaviour, and product use/consumption behaviour. In other words, we are forced to look at *predictive validity* instead of criterion validity.

The best way of evaluating the predictive validity of laddering results would be to use them to predict some actual behaviour. The literature is not replete with such studies. This is not so surprising, as there is also a lack of theory specifying how means-end chain estimates interact with situational factors to determine behaviour. However, we believe that it is possible to formulate a number of criteria that laddering procedures should meet in order to make predictive validity more likely.

Fig. 1 specifies a model of how a respondent's cognitive structure, along with the cognitive processes going on in the data collection and in the situation where the behaviour is to be predicted –the *target situation*, influence the predictive validity of the results obtained by the data collection. It also shows other factors which affect the predictive validity of the data collected with regard to the target situation: the researcher's cognitive structures and processes, and the algorithm used for data reduction. We can use this model to discuss factors that affect the predictive validity of laddering data. To this end, we will now formulate four criteria for an instrument to measure consumers' cognitive structures. The more an instrument fulfills these four criteria, the higher will be the predictive validity of the data.

2.3. Four criteria

1. The raw data should be a result more of the respondent's cognitive structures and processes than of the researcher's cognitive structures and processes.

This may sound trivial, but it is not. Any data collection method which does not allow the respondent to use some kind of natural speech is prone to violate this criterion. Measuring cognitive structures by Fishbein-type rating scales violates it, because the cognitive categories are specified in advance, and not formulated by the respondent (K.G. Grunert, 1989, 1990). The same goes for the card-sorting tasks used in some studies (e.g., Hirschman and Douglas, 1981; Valette-Florence and Rapacchi, 1990). Only when, based on previous research, there is very strong evidence of high similarity of the cognitive categories used by respondent and researcher will that type of closed questioning not violate this first criterion. Hence, the first criterion calls for open methods, in which each respondent can relate her/his own cognitive categories.

2. The data collection should not involve strategic processes not typical of the target situation.

This refers to the problem that predictive ability is impeded when the respondent uses different strategic processes in the data collection and in the target situation. Basically, there are two ways of solving this problem: (1) One can make sure that the strategic processes in the data collection situation are very similar to those one expects in the target situation, or (2) one tries to find a data collection situation with minimal strategic processing.

3. Coding should preferably be based on cognitive categories widely shared among both consumers, researchers, and users of research results, and not on the researcher's idiosyncratic cognitive categories.

Coding means grouping different answers together which, for the purpose of the following analysis, are regarded as similar. By necessity, such grouping is based on the researcher's estimate of the semantic distance between the various answers, and hence on her/his cognitive structures. This gives the researcher(s) considerable influence on the results. Again, there are basically two avenues of resolution: (1) More formalized procedures for coding answers, and/or (2) eliciting data from the respondent that are redundant, since this makes it easier to rule out ambiguities in interpretation. Both measures will help create a common understanding of the categories used between respondents, researchers, and potential users of the results.

4. The algorithm used for data reduction (aggregation) should be based on theory about cognitive structure and processes.

It should be clear what aggregation is supposed to do: summarize main aspects of the cognitive structures of a group of respondents, or derive an estimate of a collectively shared cognitive structure. In the first case, the aggregation procedure should be based on some theoretical explanation of which are the 'main' aspects of cognitive structures. In the second case, the aggregation procedure should be based on some theoretical explanation of which part of the data should enter the estimation and which part of the data is regarded as measurement error. The result of the aggregation procedure should *not* be influenced by technical aspects of the procedure which are not founded by or even in contradiction to the underlying theory, like arbitrary cutoff levels or a preference for direct rather than indirect paths in a network.

3. An examination of the collection and analysis of laddering data

In the following, we will discuss the extent to which the laddering method for measuring cognitive structures lives up to the four criteria presented in the preceding section. The data used for illustration are from a research project on determinants of the demand for organic food products (S.C. Grunert, 1993; S.C. Grunert and Kristensen, 1992). As part of that project, 29 respondents participated in laddering interviews with regard to bread. An organic and a conventional variety were presented to the respondent, who then was asked to identify differences between them. The resulting attributes were taken as starting points for the laddering process. The data processing was done using the program developed by Reynolds and Gengler [1].

To get a better understanding of the cognitive processes occurring during the data collection situation, the respondent was, after the laddering task, asked for her/his own interpretation of the task and any comments about the task. In addition, the interviewers were asked to write a comment on their impression of the laddering task, and were in turn interviewed by the authors about some of the processes occurring on the interviewer side during the interviewing task.

The validity of concurrent or post-task verbalizations of cognitive processes is in itself a ques-

[1] We would like to thank Tom Reynolds and Charles Gengler for making this program available to us.

tion of considerable debate (Ericsson and Simon, 1980, Ericsson and Simon, 1984), which will not be taken up here. We believe that respondents' comments on how they experienced a data collection technique can shed some light on the cognitive processes that occur during data collection and can give helpful hints about validity threats in the measurement process, which would not be available without such retrospective questioning.

3.1. Cognitive processes during data collection: Criteria 1 and 2

We start by discussing some of the cognitive processes which may occur during the data collection situation, in order to evaluate how well the laddering method meets criteria 1 and 2 specified above. We can immediately acknowledge that laddering allows the respondent to answer freely, in their own words –freely generate cognitive categories, thus meeting criterion 1. Criterion 2, however, requires some discussion.

Which kind of cognitive processes can be expected in a laddering interview? The basic assumption of means-end theory is that cognitive structures are hierarchical: that is, cognitive categories are ordered by level of abstraction. It is further assumed that the excerpts from cognitive structure to be measured at the individual level are of the linear type depicted in Fig. 2a. That is, we expect a sequence of cognitive categories with increasing levels of abstraction connected by one-to-one associations. However, this linear structure must be retrieved from some larger network,

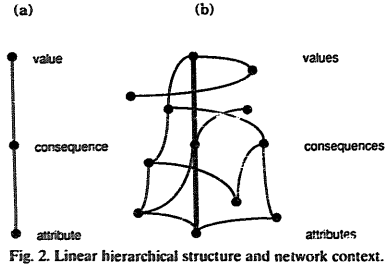


Fig. 2. Linear hierarchical structure and network context.

as in Fig. 2b. The laddering interview is intended to infer a linear hierarchical structure from a potentially much larger cognitive network.

Retrieval processes in a laddering interview can conveniently be explained by spreading activation theory (Anderson, 1983a, Anderson, 1983b; Anderson and Pirolli, 1984; Collins and Loftus, 1975), which has successfully been applied to explain retrieval in other experimental situations (e.g., Gillund and Shiffrin, 1984; Raaijmakers and Shiffrin, 1980; Strube, 1984). According to spreading activation theory, a cognitive category is retrieved (made conscious) whenever its activation surpasses a threshold. A cognitive category can be activated either because of external stimulation, because of strategic processes, or because activation spreads throughout the cognitive network from one category to the next. In that course of spreading activation, activation is lost inversely proportional to the strength of the association between the cognitive categories. The spreading of activation is assumed to be a fast, parallel, automatic process.

If we explain the retrieval processes during data collection by spreading activation theory, then an assumption about the cognitive processes occurring during laddering could be as follows: at each step in the laddering process, the interviewer's repeated question "why do you..." or "why is it important to you..." activates the cognitive category which the respondent has named last. Activation spreads from this category throughout the network, causing retrieval of additional categories, if the associations between the categories are strong enough. Presumably, the respondent would then answer with the category that has received the highest activation. This should be the concept with the strongest association to the category named last. The laddering task possibly has a bias for a certain kind of association, because causal associations (category A leads to category B) tend to be activated. At some point no strong associations exist between the category last named and any category not yet named. Then the respondent will be unable to answer, and the laddering would stop.

If the respondent's retrieval processes in the data collection would work in this way, data col-

lection would be mainly determined by automatic processes, and criterion 2 would be met. However, it seems plausible that deviations from this "ideal" laddering process can occur, especially when the cognitive structure with regard to the product in question is either especially weak or especially sophisticated.

When the cognitive structure is especially *weak* –there are few and weak associations between cognitive categories –the respondent may soon be unable to retrieve additional categories in answering questions which probe for more abstract categories. There is ample research demonstrating that in such situations (Gruenewald and Lockhead, 1980; Strube, 1984) the respondent will make a new attempt at retrieval by using a different *strategic perspective*. In terms of spreading activation theory, the respondent might initiate a new retrieval sequence based on a new pattern of starting activations. The closer one moves towards the abstract levels in the laddering process, the more likely such strategic processes are.

An example will make this clear. When a respondent has named "being healthier" as a consequence of eating organic bread, and the interviewer asks "why is it important to you to be healthier?", there may be no immediate answer. In trying to find an answer, the respondent can use different strategic perspectives. S/he can go through recent illnesses, and how life would have been without them. S/he can consider, whether s/he actually wants to be healthier, and how health may compete with other higher order values. S/he can try to imagine future life situations, where good health would be especially important, like going on a ski holiday or getting through a week of stressful worklife. In each case, quite different cognitive categories may be retrieved.

The stronger the impact of strategic perspectives on the answers given by the respondent, or, put another way, the more the task changes from a retrieval task to a problem solving task, the more criterion 2 is violated. In this case, the predictive validity of the results is likely to be lower, because it becomes increasingly unlikely that the strategic processes in the data collection situation will resemble those in a consumption situation.

In the post-laddering interview, several respondents mentioned that the laddering task led them to a considerable amount of strategic processing, especially when they got to the more abstract levels. They said that the laddering task made them think of connections they had never thought about before, and that they gained new insights during the process. This seems to indicate that the laddering task actually *changed* their cognitive structure.

One could argue that these associations were present before, but had not been retrieved into consciousness before the laddering interview. That is, the laddering procedure was successful in making explicit associations which seldom become conscious. This interpretation is impossible to refute. If we accept, however, that learning associations always require conscious awareness of the pair of elements to be learned (a view which is widely accepted, see the review by Hoffmann, 1993 [2]), then retrieval of a pre-existing association should be accompanied by a feeling of recognition, and not of surprise.

Respondents also said that they felt some pressure to come up as high as possible in the ladder, and that they felt silly when they stopped the ladder at the concrete levels. Some actually said that they thought the interviewer should

[2] An argument against the assumption that learning presupposes conscious perception could be that learning of associations can demonstrably be explained by classical conditioning, and that learning by conditioning is assumed to be an automatic process. However, to which extent the learning which occurs during conditioning involves conscious attention to the contingency of the stimuli or not is not usually investigated in conditioning studies (Allen and Madden, 1985). Of course, from a behaviouristic point of view, the question whether learning requires consciousness or not is utterly irrelevant, and hence does not require investigation. If, however, one wants to rule out the occurrence of unconscious learning, then only learning by conditioning *without conscious awareness of stimulus contingency* would invalidate the argument. Allen and Janiszewski (1989) have demonstrated in two experiments, however, that *contingency awareness* is a prerequisite for conditioning to work (see also Insko and Oakes, 1966; Petty and Cacioppo, 1981).

press them to give more answers as long as possible, because it gives a more interesting experience. Other respondents said that, when they had reached the value level, they had a feeling of having come "up too far".

Obviously, the interviewer has considerable influence on the amount of strategic processing occurring in this way. For instance, the interviewer can try to guide the respondent into answering from a specific situational context. When the respondent has no immediate answer to the question "why is it important to you to be healthier?", the interviewer may try to help by saying "have you recently experienced situations where you wanted to be healthier?" This suggests that the respondent uses a strategic perspective aimed at episodic (rather than semantic) memory. By providing a more or less relaxed interview atmosphere, automatic retrieval may be facilitated or inhibited. Finally, it is of course the interviewer's decision when to stop probing. All interviewers perceived stopping to be a very difficult decision –for example, should one accept an answer as "health is simply important to me" as final?

When the cognitive structure is especially *sophisticated*, with many strong associations, some other problems may result. One retrieval process may result in the retrieval of *several* cognitive categories, at the same level of abstraction. As an example, the probe "why is good taste important to you?" may lead to the retrieval of three answers: a) "it makes me feel relaxed and joyful", b) "my family will be pleased", and c) "it shows that I am a good housekeeper". How this affects the interview will again depend on how the interviewer reacts. When conducted in a flexible way, the interviewer may record the various answers and continue to ladder from each answer one after the other. If the interviewer insists on pursuing one ladder and urges the respondent to concentrate on one answer, the additional categories retrieved may linger in working memory and interfere with later answers to further probes, resulting in considerable strategic processing on both the respondent and the interviewer side. The further probe "why is it important for you to please your family" may then result in an answer like "my family likes it when I am a good housekeeper", which may be regarded as an attempt, from the respondent's side, to pick up some loose ends in his/her working memory and, at the same time, to come up with an answer that suits the interviewer.

Criterion 2 is thus likely to be violated when the respondent's cognitive structure with regard to the product area in question is either quite weak or quite elaborate. The way the interviewer conducts the interview will have an impact on how serious the violations are. Strategic processes, being serial in nature, are time-consuming and will therefore be noticeable to the attentive interviewer as pauses, breaks, unfinished sentences, etc. To minimize unwanted strategic processing, the interviewer should then end this line of questioning, thus avoiding ad-hoc constructions of new associations. Likewise, by allowing 'forked' answers (as in the example about good taste), without interrupting the natural flow of speech from the respondent, the interviewer can avoid strategic processes due to forcing a structure on answers which the respondent regards as unnatural.

We may designate this type of interview, where the natural flow of speech of the respondent is restricted as little as possible, as 'soft' laddering. In contrast, 'hard' laddering refers to interviews and data collection techniques where the respondent is forced to produce ladders one by one, and to give answers in such a way that the sequence of the answers reflects increasing levels of abstraction. Data collection techniques that do not involve personal interviews at all, like self-administered questionnaires (Walker and Olson, 1991; Young and Feigin, 1975) and computerized data collection devices, are all examples of 'hard' laddering. Such techniques make it impossible to detect strategic processing by the respondent.

The 'soft' approach is potentially better in handling the types of problems on the respondent side caused by weak or elaborate cognitive structures. On the other hand, it obviously increases the extent of cognitive processing on the interviewer side, which may introduce new biases. The interviewer must try to make sense of the answers, and relate them to the means-end model. This will be unproblematic when the respondent

produces "ideal" laddering responses and retrieves items of semantic knowledge, one by one, at increasing levels of abstraction. However, as noted, respondents often deviate from this. For example, people may retrieve episodic instead of semantic information, which usually means that they start telling little stories ("last time I bought this, I noticed..."). People may give an answer and immediately start elaborating it ("I like bread with wrinkles. Not wrinkles generally, but, you know, the type which..."). People may jump back and forth between the levels of abstraction. All this requires cognitive processing on the interviewer side in order to distil ladders: generalize from the episodic to the semantic, simplify the elaborate, and sort out levels of abstraction. The more this occurs, the more influence the interviewer will have on the results. 'Hard' laddering is an attempt to avoid this. The choice between 'hard' and 'soft' laddering can therefore be viewed as balancing the unwanted effect of strategic processing by the respondent and the unwanted effect of additional interviewer involvement. We would like to propose that 'softer' methods are more appropriate whenever we can expect the cognitive structure of the respondent to be weak, due to low involvement and/or little experience with the product, and whenever the cognitive structure is very elaborate, due to high involvement and much experience, such as when the respondent is an expert in the respective area. According to the same argument, 'harder' methods may be more useful in cases of average involvement and experience.

3.2. Cognitive processes during coding: Criterion 3

As noted in the preceding section, not only the respondent's cognitive processes, but also the interviewer's or researcher's cognitive processes during the data collection situation have an impact on the results and their validity. But the researcher's cognitive structure really starts to have an impact on the results of the study during the coding process.

Coding means grouping different answers together which, for the purpose of the following analysis, will be regarded as equal. In analysing laddering data, coding is a prerequisite for calculating implication matrices and hierarchical value maps. Also, each category is usually characterized as being either an attribute, or a consequence, or a value. Our criterion 3 requires that coding should be based on cognitive categories widely shared among consumers, researchers, and users of research results, and not on the researcher's idiosyncratic cognitive categories. This means that different researchers should perform the coding based on a common understanding of the meaning of the coded material, and that the meaning the researcher attaches to the material and the meaning the respondent originally attached to the material should not be too divergent. This somewhat loose formulation reflects the fact that there will never be a complete congruence of meaning between respondent and researcher and/or different researchers. The deviations are a matter of degree. But everybody will probably agree that large deviations of that kind are a major validity threat.

We discuss three problem areas in coding, which can impact the extent to which such kinds of deviations will occur. They are the distinction between attributes, consequences, and values; the indexicality problem; and the problem of finding the 'right' level(s) of abstraction.

The *distinction between attributes, consequences, and values* should, of course, be based on a conceptual definition of these terms. The laddering literature is surprisingly void of such definitions. In practice, many borderline cases turn up. When the respondent says "healthy" –is this an attribute or a consequence? "The bread is healthy" probably designates an (abstract) attribute. "I will be healthy when I eat this bread" seems to be a consequence. But, isn't "health" a value? Making such categorizations in a uniform way is heavily dependent on the availability of context information, i.e., on the extent to which a term like *healthy* is embedded in a redundant flow of natural speech. How much context information is available will again depend on the way the laddering interview is executed. Laddering done by self-administered questionnaires or by computer (the 'hard' form) usually provides very little context information. In laddering tasks where the interviewer records the answers as

notes in ladder schemes, the context information is available to the interviewer only and not in subsequent coding. Only when the interviews are taped and transcribed is the full context available in coding.

The categorization of a term like *healthy* as an attribute or a consequence may not be decisive for grasping the meaning the respondent wants to express. However, the need for context to attach meaning to an answer refers actually to the much more general problem qualitative researchers call *indexicality*: For the researcher, it is possible to understand, or make sense of a respondent's answer, only by relating it to that respondent's individual background. Answers must be interpreted relative to the respondent's background, experience, career, her/his interpretation of the data collection situation, etc. (Hoffmann-Riem, 1980; Hopf, 1978; Küchler, 1981; Schütze et al., 1973). When taken very seriously, the call for indexicality prohibits all coding and generalisation, because every respondent has to be regarded as unique (K.G. Grunert, 1990; Mohler, 1981). However, it is evident that the indexicality problem is more serious with some research topics than with others. Most people are comfortable with ascertaining sex and age by closed questions, because the cognitive categories used by respondent and researcher can safely be assumed to be the same. But the less one knows in advance about how respondents think about the topic to be researched, i.e., the less one knows about the product attributes, consequences, and values likely to be used by them to attach meaning to a certain product category, the more serious the indexicality problem. Thus it is more important to devise a data collection method that helps the researcher understand the meaning in the respondent's answers. Again, the more context information a data collection device supplies, the easier this process of assigning meaning will be. This is why methods employing natural speech, like focus groups and depth interviews, are better in meeting the call for indexicality than methods which artificially limit the respondent's answers to prespecified categories, like in a closed questionnaire, or to single words, like in a word association task. As will be obvious from the above, the indexicality of laddering data depends very much on the way a laddering interview is conducted. A 'hard' laddering interview, where the interviewer or an impersonal data collection device force the respondent to answer in sequential ladders, will be lower in indexicality. In contrast, a 'soft' laddering interview, where the means-end idea is used to structure data in an otherwise free-flowing, natural speech, will be higher in indexicality. The less previous knowledge we have about respondents' cognitive categories, the more appropriate 'soft' methods become.

The third problem in coding is to find the "right" *level of abstraction*. The difference between two answers is rarely purely lexical. To define them as synonyms and group them into the same category, the category has to be at a more abstract level than the answers themselves. For example, "excellent taste" and "pretty good taste" may be both sorted into "good taste". "Good taste" and "bad taste" may both be sorted into "taste". "No chemicals" and "no preservatives" may be sorted into "only natural ingredients". Such codings appear intuitively unproblematic, even though they obviously lead to a loss of information.

If the coding stops at that level, the resulting number of categories will still be large (easily 40–50 concepts, as this and other studies have shown). The implication matrix will be correspondingly large, the cell frequencies will be low, and it will not be possible to compute and draw a hierarchical value map, destroying the most appealing device of the laddering technique. Thus, a more radical coding is typically required to reduce the data (Gutman, 1991). This means, usually, that the level of abstraction for each category has to be raised considerably. "No artificial colours", "no preservatives", "more minerals", "better ingredients", "freshly milled flour" all become "ingredients". Many will still find this uncontroversial. It is at the consequences and value levels, the real difficulties start. Can "joy" and "not being depressed" both be coded into "well-being"? Or "have experience" and "curiosity" into "variation"? Such rather broad categories usually have to be created, if a technically manageable implication matrix is to result.

It is obvious that coding is a complicated process that gives a lot of latitude to the researcher, and much more attention should be paid to it in the typical methodology discussion than has been done up till now. In particular, the process should be made more intersubjectively accessible. Having parallel coders is of course the most common recourse used in research practice. But it may be helpful to draw on some of the experiences and tools developed within the realm of computer-assisted content analysis (e.g., K.G. Grunert and Bader, 1986; Züll and Mohler, 1992), which is concerned with finding more intersubjective ways of coding text data. The basic idea developed is that of *iterative coding*. This means that a first coding is performed, and the implications of this coding are made transparent by aids like keyword-in-context lists, leftover lists, and insertion of codes into the text data base. Based on these aids, the coding is revised, and the implications of the revised coding are analysed in the same way. This procedure continues until the coding appears satisfactory. Of course, the decision about what can be regarded as 'satisfactory' still rests to a large extent on face validity considerations and, therefore, on the judgement of the individual researcher. However, such procedures provide documentation for how the coding has proceeded, thus increasing the intersubjectivity of the process.

3.3. Algorithms for data reduction: Criterion 4

Our final criterion required that the algorithm used for data reduction (aggregation) should be based on underlying theory about cognitive structure and processes. Data reduction in analysing laddering data encompasses two main steps: *aggregation* and *condensation* [3].

Aggregation involves the step from the individual to the collective. The hierarchical value map, the main output from a laddering analysis, is a characterization of a group of respondents. There are two possible views, one modest, one ambitious, of hierarchical value maps. The modest view is that a hierarchical value map is a device that allows us to see the major results from a laddering study of a group of respondents without having to go through all the individual ladders. The more ambitious view is that the hierarchical value map is an estimate of cognitive structure for that group of respondents. While laddering literature does not take a clear stand on which view one should adopt, the more ambitious view would be in line with much previous research on estimating cognitive structures, especially within the word association paradigm (e.g., Deese, 1965; Szalay and Deese, 1978). The argument for the more ambitious view runs as follows: at the individual level, our data are not rich enough to estimate a respondent's cognitive structure. In a laddering study, the 2–3 ladders typically obtained from an individual respondent reveal some aspects of his/her cognitive structure, but they are not an estimate of the cognitive structure itself, because the cognitive structure itself is not a collection of single chains, but an interrelated net of associations. However, when we obtain ladders from a group of *homogeneous* respondents, then the set of ladders obtained from them, taken together and analysed by an appropriate algorithm, will yield an estimate of this group's cognitive structure.

The view taken should determine which operations are regarded as admissible when aggregating laddering data. In the following discussion, we assume the more ambitious view. However, the problems discussed will affect the interpretability of hierarchical value maps also when only the modest view is adopted.

We will illustrate the problems which can occur in aggregation with an example from our bread study. Fig. 3 shows two hierarchical value maps. We have divided the respondents into two groups, depending on whether they regularly buy organic food products, or whether they never or rarely buy organic food products. The Figure shows that the two hierarchical value maps for these two groups are quite different. The regular buyers of organic foods have two groups of attributes: the more physical characteristics of the

[3] The term *condensation* is used in graph theory for a network which covers the essential aspects of a larger graph (see Harary et al., 1965).

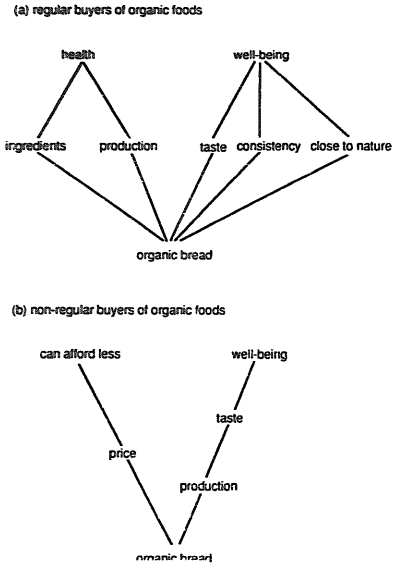


Fig. 3. Hierarchical value map for regular buyers (n = 15) and non-regular buyers (n = 16) of organic food (cutoff level = 4).

bread, namely ingredients and production method, which they associate with health. Then the more intangible attributes, like taste, consistency, and "close to nature", which are associated with well-being. The non-regular buyers of organic foods mention high price, with the consequence of being able to afford less for other purchases. They also mention production method, but it is not associated to health, but to (poor) taste.

The reader may at this point try to guess how a hierarchical value map for the two groups pooled would look. Actually, the result is still more dramatic than the differences in Fig. 3 suggest. The reasons are of a technical nature, and we need to discuss *condensation* in order to understand them.

Condensation refers to reducing the hierarchical value map to a small subset of the associations between cognitive categories that have shown up in the data. In principle, one could, based on the implication matrix, draw a map which shows all the cognitive categories which resulted from the coding process, and in which two cognitive categories are linked whenever the corresponding cell in the implication matrix has a non-zero entry. In practice, this is seldom possible or desirable. In the bread study, the coding resulted in 39 categories, yielding 741 possible links, if we disregard asymmetries. In the implication matrix, 354 cells actually had non-zero entries. In practice, one tries to find a hierarchical value map that includes the "most important" links. This is achieved by two means –first, by specifying the condition that the network has to be *nonredundant*, and secondly, by specifying a *cutoff level*.

Non-redundancy means that, if category 1 at abstraction level A is linked to category 2 at abstraction level B, which again is linked to category 3 at abstraction level C, then there should not be a direct link between categories 1 and 3, because such a link would be redundant. This assumption, which is also called the *economy-of-storage* assumption, has been debated for a long time in research on semantic memory (see Chang, 1986, for an overview). There is experimental evidence both for and against it. But it is an assumption which is used in many successful models of semantic memory, and using it in the context of means-end chain research may thus be defensible.

The assumption of non-redundancy (or economy-of-storage) first becomes problematic in the context of aggregation. If respondent I has a ladder 1–2–3, and respondent II has a ladder 1–3, then both underlying cognitive structures may, at the individual level, conform to the principle of non-redundancy. But how to aggregate the data? The way this is usually handled in creating a hierarchical value map [4] is as follows. If, *at the aggregate level*, there is a link 1–2–3 (categories 1 and 3 are linked indirectly), then there will be no link 1–3, *even if such links were observed at the individual level*. This can have

[4] Both the programs of Reynolds and Gengler and of Valette-Florence and Rapacchi use this solution.

rather astonishing results. Imagine that, out of a sample of 30 respondents, 25 have the ladder *taste –well-being*, and 5 have the ladder *taste –function better –well-being*. In the aggregate map, the whole group of respondents would then be characterized by *taste –function better –well-being*, because a direct link between *taste* and *well-being* would be redundant. The map hence communicates the erroneous impression that, for this group of respondents, bread tasting better leads to the personal consequence that they can function better, with all kinds of marketing implications. Actually, most of the respondents just said that they enjoy a good taste.

The most extreme cases, where only one respondent has named the intermediate category, while all others have not, are usually avoided by specifying a cutoff level. The cutoff level gives the minimum cell entry in the implication matrix necessary to be represented as a link in the map. Since the distribution of the cell entries is usually heavily skewed –many cell entries are very low, while only a few are high –the cutoff level is a powerful device for reducing the complexity of the map. The problem is that there are no theoretical or statistical criteria to guide the selection of the cutoff level. Thus, one usually attempts some compromise between retaining information on the one hand and creating a manageable map on the other hand.

If subsequent raises of the cutoff level would just result in maps that, to an increasing degree, are reduced to the strongest associations, this could be acceptable. However, because of the assumption of non-redundancy in the aggregated map, this is not the case. By raising or lowering the cutoff levels, intermediate categories may be deleted or added, changing the resulting map quite considerably.

Fig. 4 illustrates these problems by showing the hierarchical value map of the whole group of respondents pooled, employing the same cutoff value as in the separate value maps in Fig. 3. The result looks decidedly different. First of all, there are considerably more links and categories than a pure joining of Figs. 3a and 3b would suggest. This is because, by pooling the data, we have doubled the number of respondents, but used the same cutoff level –making it more likely, for any cell in the implication matrix, that its value becomes greater or equal to the cutoff value. Thus, we have in this map several cognitive categories and links which appeared in neither Fig. 3a nor 3b.

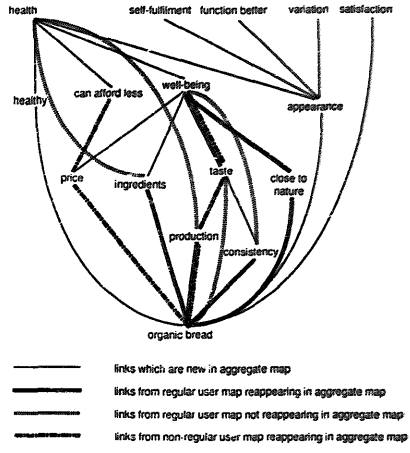


Fig. 4. Comparison of value map for pooled respondents with value map for regular and non-regular users of organic food products.

Can we locate the structure from Figs. 3a and 3b in Fig. 4? From a validity point of view this would be desirable, because otherwise the structure of the map is influenced by technical details, like the choice of a cutoff level, and the way the data are pooled. In Fig. 4, the structure from Figs. 3a and 3b has been superimposed on the map, in order to facilitate a comparison. We see that the value map of the non-regular users of organic foods, as depicted in Fig. 3b, is a subset of the new, pooled map. This is as it should be. We see that the same can not be said about the value map of the regular users of organic foods. Recall the interpretation of their structure in Fig. 3a: The regular buyers of organic foods have two groups of attributes, the more physical characteristics of bread, namely *ingredients* and *production*

method, which they associated with *health*, and the more intangible attributes, like *taste, consistency*, and *close to nature*, which are associated with *well-being*. This can no longer be seen in the pooled map. Instead of the direct link from *ingredients* to *health*, there is now a link from *ingredients* to *well-being*, and from there to *health*, changing the two-wing structure from Fig. 3a. Likewise, there is no direct link from *production* to *health*, but from *production* to *taste*, from there to *well-being*, and from there to *health*. The original distinction between the health-directed structure and the well-being-directed structure is completely lost.

The reason for this is the combined effect of pooling the data, the non-redundancy assumption, and the choice of cutoff level. By having more respondents in the analysis and hence having a relatively lower cutoff level, new links are added to the map. One of these is the link between *well-being* and *health*. The non-regular buyer group has a chain *production-taste-well-being*. This together gives an indirect link between *production* and *health*, which because of the non-redundancy assumption, eliminates the direct link between *production* and *health*, which characterized the regular buyers. The other "lost links" in Fig. 4 can be explained in a similar way.

These effects are not trivial from an application point of view. It makes a difference when advertising that the different way organic bread is produced is preferred because one expects it to give better health, as Fig. 3a suggests, or because one believes it leads to better taste, as Fig. 4 suggests.

In trying to develop better algorithms for the analysis of laddering data, the first step should be an explicit stand on what a hierarchical value map is supposed to do. If it is meant as an estimate of cognitive structure, then the next step would be to spell out clearly the assumptions made about the nature of that structure, especially about non-redundancy/economy-of-storage. The technical problem to be solved then is to aggregate only respondents whose cognitive structures can be regarded as reasonably homogeneous with regard to the product in question. This should be achieved by applying clustering methods to the laddering data before aggregation. If, on the other hand, the more modest view is adopted, i.e., a hierarchical value map is only a graphical device for the purpose of summarizing main results from a laddering study, then the non-redundancy assumption should be relaxed, and algorithms should be developed that allow redundant links. This may make the maps a little harder to read, but it may avoid misinterpretations.

4. Conclusions on the predictive validity of the laddering method

We have tried to evaluate the validity of the laddering method. This turned out to be a difficult task, partly because there is no general agreement on what laddering is supposed to do. We have adopted a classical positivistic stance here, assuming that the purpose of laddering is to measure consumers' cognitive structures with regard to certain product areas. This leads to the validity notion of classical test theory, which, in the present context, would imply that the laddering method has validity to the extent that it actually results in estimates of consumers' 'true' cognitive structures. However, 'true' cognitive structures are not accessible, since their relationship to behaviour is always mediated by cognitive processes. We therefore turn to the notion of predictive validity, namely to which extent the result of cognitive structure and cognitive processes which the laddering procedure yields can be expected to accurately predict behaviours toward the product groups in question. Such a discussion is made difficult by the lack of a theory specifying how laddering results can be used to predict behaviour; nevertheless, we proposed four criteria that will affect the predictive validity of laddering results.

1. The raw data should be a result more of the respondent's cognitive structures and processes than of the researcher's cognitive structures and processes. Laddering meets this criterion, because the respondent is free to use his/her own cognitive categories. Even though the interviewer may

have considerable influence on the course of the interview, s/he will not suggest particular attributes, consequences or values in a laddering study.

2. The data collection should not involve strategic processes atypical for the target situation. Even though the cognitive structure of the respondent may remain unchanged between the interview and the performance of the behaviour to be predicted, results from a laddering study will lack predictive validity if the cognitive processes producing answers in the interview are very different from those in the situation where behaviour is to be predicted. If the respondent has little knowledge about the product area, but the interviewer nevertheless pushes him/her far up the ladder of abstraction, the respondent may construct new associations to meet the interview demands. If the respondent has comprehensive knowledge about the product area, s/he may find it difficult to follow the hierarchical format of laddering. In both cases, strategic processing atypical for the target situation may result.

3. Coding should preferably be based on cognitive categories widely shared among consumers, researchers, and users of research results, and not on the researcher's idiosyncratic cognitive categories. In coding, the researcher has to reconstruct the meaning the respondent tried to bring across during the interview, which will be more or less difficult depending on the redundancy in the data. In addition, the researcher has to group these meanings into categories at considerably higher levels of abstraction, opening the way for considerable influence of the researcher's idiosyncratic cognitive categories.

4. The algorithm used for data reduction (aggregation) should be based on theory about cognitive structure and processes. Hierarchical value maps, the typical main output of a laddering study, may be viewed in two ways: as an estimate of cognitive structure, or just as a device to highlight major results from analysing a group of respondents. If it is viewed as an estimate of cognitive structure, only respondents previously classified as reasonably homogeneous should enter the map, as it will be misleading otherwise. If it is used only as a device to highlight major results from a group of respondents, the non-redundancy assumption used in constructing these maps leads to misleading results, because such an assumption makes sense at the individual level, but not if cognitive structures from non-homogeneous respondents are aggregated.

5. Implications for an improved laddering methodology

We have thus identified possible threats to validity with respect to criteria 2, 3 and 4. They suggest improvements in data collection, coding, and analysis.

Data collection. We introduced the distinction between 'hard' and 'soft' laddering. 'Hard' laddering forces the respondent to produce ladders one by one, whereas 'soft' laddering allows forks, loops, and blind alleys. 'Soft' laddering presupposes a trained interviewer, whereas 'hard' laddering can be done by computer or by self-administered questionnaires. We suggest that 'soft' laddering is more appropriate whenever unwanted strategic processes on the respondent side are judged to be likely –when the degree of knowledge of the respondent about the product area is either low or high –because the interviewer can detect such processes and steer the interview accordingly. Also, 'soft' laddering can be expected to yield more redundant data, which makes the reconstruction of meaning in the coding phase easier. When unwanted strategic processes are expected to be less likely, and when it is judged that the area to be researched is so well-known in advance that the reconstruction of meaning will be unproblematic, 'hard' laddering has the advantage of minimizing interviewer influence.

Coding. The coding procedure should be made more transparent in order to obtain a higher degree of intersubjectivity. Ideally, a tree diagram should be available to document the whole coding process from the raw data to the coded data. The process of arriving at the categories can be made more intersubjectively accessible by using

methods from computer-assisted content analysis, like keyword-in-context lists, leftover lists, and insertion of codes into the raw text.

Data analysis. The present algorithms should be supplemented in two ways. First, there should be clustering options that, based on the raw data, cluster a set of respondents into subgroups of reasonable homogeneity before further aggregation/condensation. Secondly, there should be different options for the redundancy in hierarchical value maps.

In conclusion, we wish to emphasize that the problems discussed here are not simply a matter of method improvement. They underline that the conceptual status of means-end chains and laddering is unclear. More clarification would be desirable on the following questions: What does laddering measure? What are the results to be used for? If means-end chains are to have an explanatory/predictive function, what kind of behaviour is to be explained/predicted, and in which way? What exactly is a hierarchical value map, and how should it be used? Answers to these questions would require the development of theory that integrates means-end chains into a larger cognitive model of consumer behaviour, a model that also deals with the links between cognitive structure and behaviour which have typically been dealt with within a framework inspired by the theory of reasoned action.

References

Allen, C.T. and C.A. Janiszewski, 1989. Assessing the role of contingency awareness in attitudinal conditioning with implications for advertising research. Journal of Marketing Research 26, 30–43.

Allen, C.T. and T.J. Madden, 1985. A closer look at classical conditioning. Journal of Consumer Research 12, 301–315.

Anderson, J.R., 1983a. A spreading activation theory of memory. Journal of Verbal Learning and Verbal Behavior 22, 261–295.

Anderson, J.R., 1983b. The architecture of cognition. Cambridge, MA: Harvard University Press.

Anderson, J.R. and P.L. Pirolli, 1984. Spread of activation. Journal of Experimental Psychology: Learning, Memory and Cognition 10, 791–798.

Asselbergs, P., 1989. Competitive advertising –New developments in qualitative positioning research: Meaning structure analysis. Rotterdam: IPM.

Chang, T.M., 1986. Semantic memory: Facts and models. Psychological Bulletin 99, 199–220.

Collins, A.M. and E.F. Loftus, 1975. A spreading-activation theory of semantic processing. Psychological Review 82, 407–428.

Deese, J., 1965. The structure of associations in language and thought. Baltimore: Johns Hopkins.

Dichter, E., 1960. The strategy of desire. New York: Doubleday.

Ericsson, K.A. and H.A. Simon, 1980. Verbal reports as data. Psychological Review 87, 215–252.

Ericsson, K.A. and H.A. Simon, 1984. Protocol analysis: Verbal reports as data. Cambridge, MA: MIT-Press.

Gillund, G. and R.M. Shiffrin, 1984. A retrieval model for both recognition and recall. Psychological Review 91, 1–67.

Gruenewald, P.J. and G.R. Lockhead, 1980. The free recall of category examples. Journal of Experimental Psychology: Human Learning and Memory 6, 225–240.

Grunert, K.G., 1989. Another attitude on multi attribute attitude theories. In: K.G. Grunert and F. Ölander (eds.), Understanding economic behaviour, 213–230. Dordrecht: Kluwer.

Grunert, K.G., 1990. Kognitive Strukturen in der Konsumforschung. Heidelberg: Physica.

Grunert, K.G., 1991. Automatic and strategic processes in advertising effects: Attention, perception and learning. Paper presented at the Annual Conference of the European Marketing Academy, Dublin, May 21–23, 1991.

Grunert, K.G. and M. Bader, 1986. Die Weiterverarbeitung qualitativer Daten durch computerunterstützte Inhaltsanalyse. Marketing –ZFP 4, 238–247.

Grunert, S.C., 1993. Green consumerism in Denmark: Some evidence from the ØKO foods-project. Der Markt 32, 140–151.

Grunert, S.C. and K. Kristensen, 1992. Den danske forbruger og økologiske fødevarer. Århus: Handelshøjskolen i Århus, Institut for Informations-behandling.

Gutman, J., 1982. A means-end chain model based on consumer categorization processes. Journal of Marketing 46, 60–72.

Gutman, J., 1991. Exploring the nature of linkages between consequences and values. Journal of Business Research 22, 143–148.

Harary, F., R.Z. Norman and D. Cartwright, 1965. Structural models: An introduction to the theory of directed graphs. New York: Wiley.

Hirschman, E.C. and S.P. Douglas, 1981. Hierarchical cognitive content: Towards a measurement methodology. In: K.B. Monroe (ed.), Advances in consumer research, vol. 8, 100–105. Chicago: Assocation for Consumer Research.

Hoffmann, J., 1993. Unbewußtes Lernen –eine besondere Lernform? Psychologische Rundschau 44, 75–89.

Hoffmann-Riem, C., 1980. Die Sozialforschung einer interpretativen Soziologie –Der Datengewinn. Kölner Zeitschrift für Soziologie und Sozialpsychologie 32, 339–372.

Hopf, C., 1978. Die Pseudo-Exploration –Überlegungen zur Technik qualitativer Interviews in der Sozialforschung. Zeitschrift für Soziologie 7, 97–115.

Insko, C.A. and W.F. Oakes, 1966. Awareness and the "conditioning" of attitudes. Journal of Personality and Social Psychology 4, 487–496.

Jolly, J.P., T.J. Reynolds and J.W. Slocum, 1988. Application of the means-end theory for understanding the cognitive bases of performance appraisal. Organizational Behavior and Human Decision Processes 41, 153–179.

Küchler, M., 1981. Kontext –Eine vernachlässigte Dimension empirischer Sozialforschung. In: J. Matthes (ed.), Lebenswelt und soziale Probleme, 344–354. Frankfurt: Campus.

Mohler, P.P., 1981. Zur Pragmatik qualitativer und quantitativer Sozialforschung. Kölner Zeitschrift für Soziologie und Sozialpsychologie 33, 716–734.

Olson, J.C., 1989. Theoretical foundations of means-end chains. Werbeforschung and Praxis 5, 174–178.

Olson, J.C. and T.J. Reynolds, 1983. Understanding consumers cognitive structures: Implications for advertising strategy. In: L. Percy and A.G. Woodside (eds.), Advertising and consumer psychology, 77–90. Lexington, MA: Lexington.

Peter, J.P. and J.C. Olson, 1990. Consumer behavior, 2nd ed. Homewood, IL: Irwin.

Petty, R.E. and J.T. Cacioppo, 1981. Attitudes and persuasion: Classic and contemporary approaches. Dubuque, IO: Wm.C.Brown.

Raaijmakers, J.G.W. and R.M. Shiffrin, 1980. SAM: A theory of probabilistic search of associative memory. In: G.H. Bower (ed.), The psychology of learning and motivation, Vol. 14, 207–262. New York: Academic Press.

Reynolds, T.J. and J. Gutman, 1988. Laddering theory, methods, analysis and interpretation. Journal of Advertising Research, 11–31.

Schneider, W. and R.M. Shiffrin, 1977. Controlled and automatic human information processing: I. Detection, search and attention. Psychological Review 84, 1–66.

Schütze, F., W. Meinefeld, W. Springer and A. Weymann, 1973. Grundlagentheoretische Voraussetzungen methodisch kontrollierten Fremdverstehens. In: Arbeitsgruppe Bielefelder Soziologen (ed.), Alltagswissen, Interaktion und gesellschaftliche Wirklichkeit, 433–495. Reinbek: Rowohlt.

Strube, G., 1984. Assoziation. Heidelberg: Springer.

Szalay, L.B. and J. Deese, 1978. Subjective meaning and culture: An assessment through word associations. Hillsdale, NJ: Erlbaum.

Valette-Florence, P. and B. Rapacchi, 1990. A cross-cultural means-end chain analysis of perfume purchases. In: N.E. Synodinos et al. (eds.), Proceedings of the third symposium on cross-cultural consumer and business studies, 161–172. Honolulu, HI: University of Hawaii.

Walker, B.A. and J.C. Olson, 1991. Means-end chains: Connecting products with self. Journal of Business Research 22, 111–118.

Young, S. and B. Feigin, 1975. Using the Benefit Chain for improved strategy formulation. Journal of Marketing 39, 72–74.

Züll, C. and M., P.Ph. Mohler, 1992. Textanalyse. Opladen: Westdeutscher.